

# Pathophysiology of Arrhythmias

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# Session Objectives

- Describe basic mechanisms of arrhythmia formation
- Explain the formation of reentry, triggered activity, and abnormal automaticity
- Identify common factors affecting arrhythmia formation
- Interpret common clinical arrhythmias and their underlying mechanisms
- Identify common cardiac channelopathies and analyze their mechanisms for arrhythmia formation

# Definition of Arrhythmias

=not normal sinus rhythm

## Normal heart beat:

1. Origin: Sinoatrial-SA node or the sinus node (SN)
2. Rate: 60-100bpm
3. Rhythm: regular
4. Conduction sequence  
SN-Atria-AVN-His bundle (supraventricular)-  
Bundle branches-Purkinje system-Ventricular myocytes

Any deviation from normal heart beat is arrhythmia

Note that some arrhythmias could be physiological

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➤ By heart rate

- Gaztanaga et al, Rev Esp Cardiol. 2012;65(2):174–185

# Arrhythmia classifications (2)

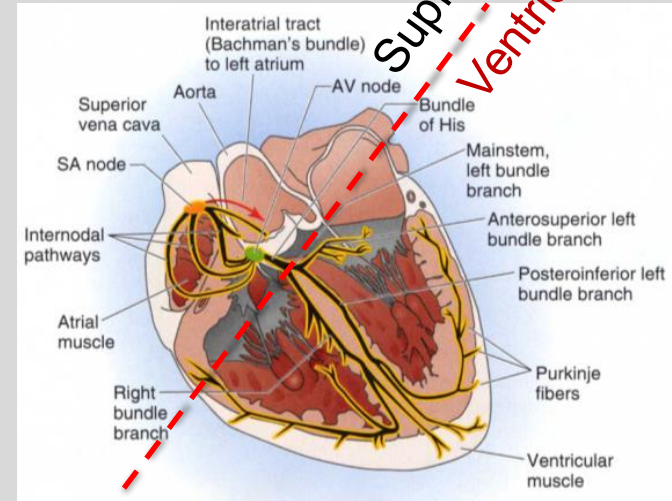
## ➤ For tachy-arrhythmias

### 1. Supraventricular arrhythmias

- originating from a site at His bundle or above
- with normal activation of ventricles (unless there is bundle branch block)
- typically with normal QRS (narrow)

### 2. Ventricular arrhythmias

- below His bundle
- abnormal ventricular activation
- widened QRS



# Arrhythmia mechanisms

## 1. Abnormal Impulse Initiation

### 1). Abnormal automaticity

- a. Altered normal automaticity
- b. Abnormal automaticity

### 2). Triggered activity

- a. Early afterdepolarization  
(EAD)
- b. Delayed afterdepolarization  
(DAD)

## 2. Abnormal Impulse Conduction

### 1). Conduction block

common sites: SN-atrial,  
AV node, His bundle &  
bundle branches

### 2). Reentry

- a. Anatomic reentry
- b. Functional reentry

# Altered Normal Automaticity

Increased (too fast,  
>100bpm):

**Sinus tachycardia**-usually secondary, could be physiological (e.g., exercise), also seen in disease states (fever, thyrotoxicosis, etc.)

**Inappropriate sinus tachycardia**

Decreased (too slow,  
<60bpm):

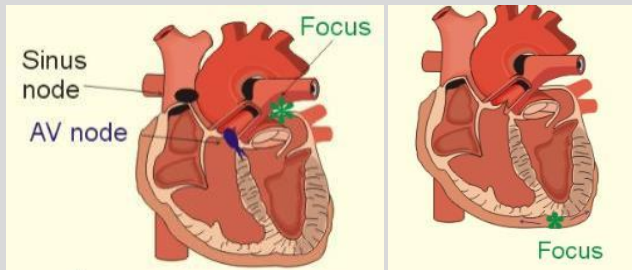
**Sinus bradycardia**-could be physiological (at rest), or in diseases (such as Sick Sinus Syndrome, SSS)

# Abnormal Automaticity

## Non-pacemaker myocyte (atrial & ventricular)

In disease state, could  
gain pacemaker activity

Atrial arrhythmias  
Ventricular arrhythmias



➤ Common factors promoting  
abnormal automaticity:  
hypoxia, ischemia, inflammation, etc.

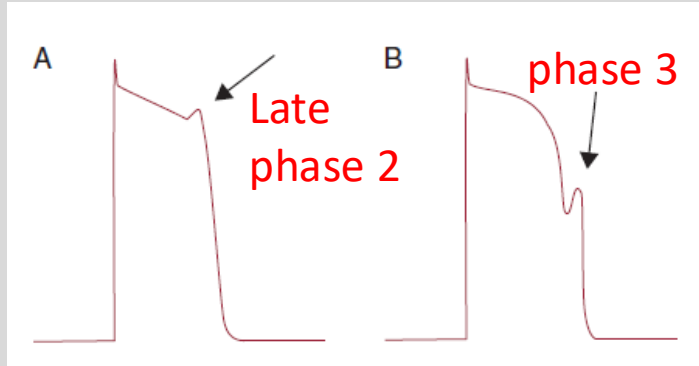
cause cell injury → membrane potential  
increases (more positive) → inactivates  
fast Na<sup>+</sup> channels

could become slow response cells  
(excites by I<sub>Ca-L</sub>)

# Triggered Activity

(triggered by the previous AP)

## 1. Early afterdepolarization (EAD)

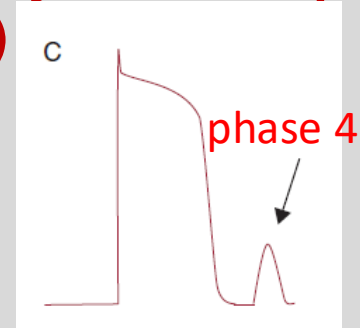


- Linked to prolonged AP duration: Long QT syndrome, hypokalemia, etc.
- due to re-opening of L-type  $\text{Ca}^{2+}$  Channel
- Risk  $\uparrow$  at slower heart rate (bradycardia prolongs AP duration)

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## 2. Delayed afterdepolarization (DAD)



- Linked to  $\text{Ca}^{2+}$  overload: Digitalis toxicity, reperfusion-induced arrhythmia, CPVT (Catecholaminergic Polymorphic Ventricular Tachycardia) etc.
- Due to  $\uparrow$  Na/Ca exchanger current
- Risk increases at faster heart rate

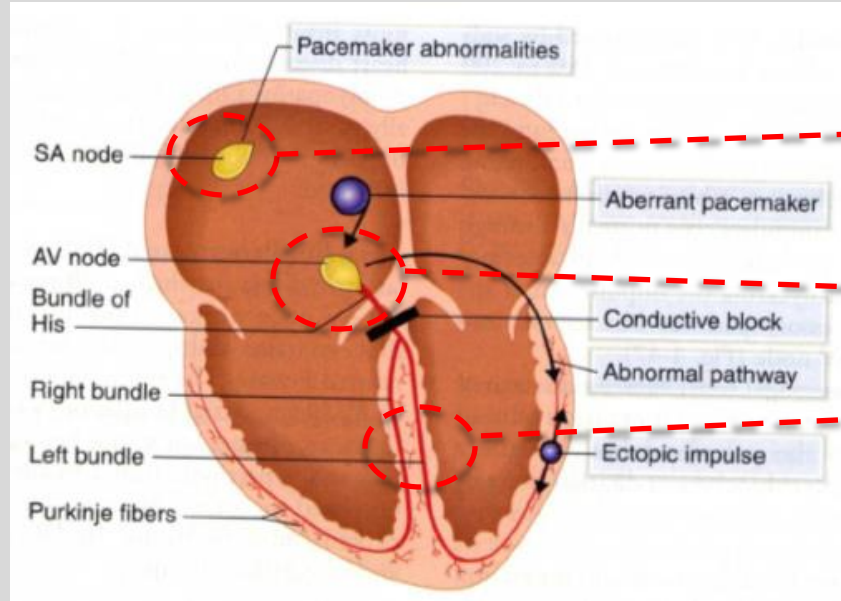
# Abnormal Impulse Conduction (1)

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## Conduction block (causes bradyarrhythmias)

### Common conduction blocks:



Sino-atrial exit block  
(sinus bradycardia or arrest)

AV (atrioventricular) block

Left or Right bundle branch  
block (BBB)

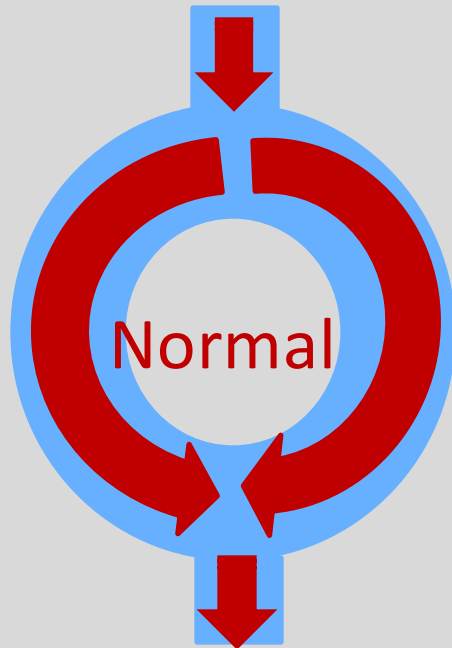
# Abnormal Impulse Conduction (2)

## Reentry

Normally a heart beat from the sinus node will excite the whole heart once and die out.

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**Definition of Reentry-**  
if the same impulse re-  
excites the heart again

# Formation of reentry

A1 | A1 | A1

CV: slow

Different EP properties

ERP: short

CV: **Fast**  
(Conduction velocity)

ERP: **Long**  
(effective refractory period)

Normal beats A1 (slow, regular)

A1 | A1 | **A2**

Premature beat

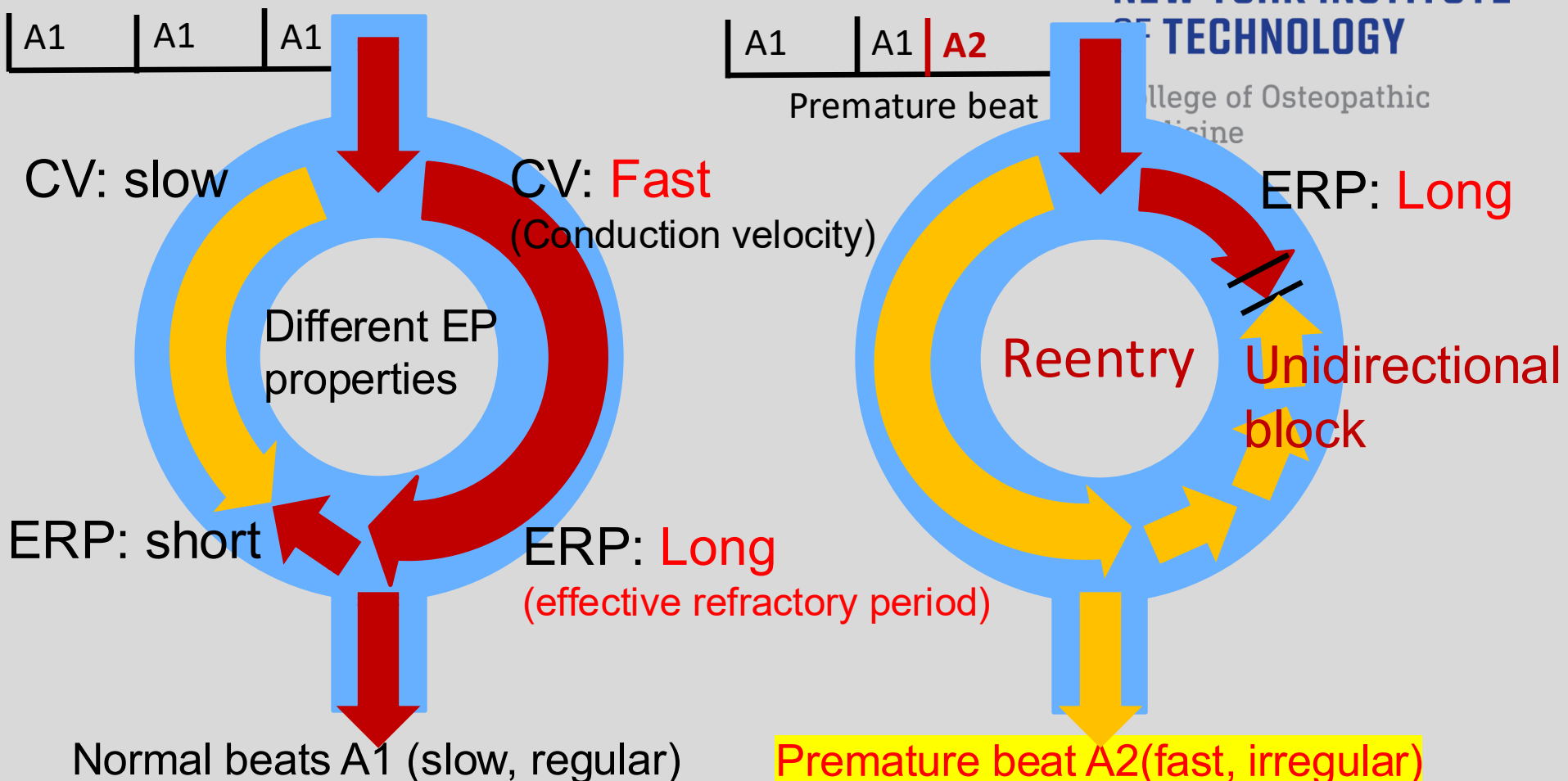
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ERP: **Long**

Reentry  
Unidirectional  
block

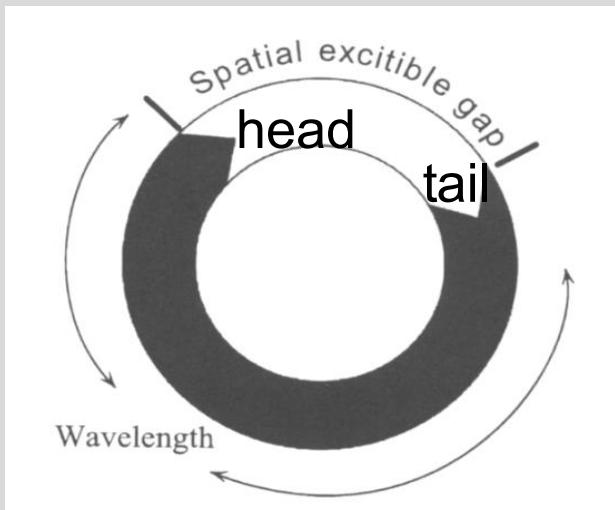
Premature beat A2 (fast, irregular)



# Requirements to form reentry

- 1. unidirectional block**
- 2. slow conduction**
- 3. recovery of previously excited proximal tissue**

# Wavelength and excitable gap in reentry



## Reentry

$WL < \text{Length of reentry circuit}$   
(The head never catches up the tail)

$\uparrow WL \quad \downarrow \text{reentry risk}$

$\downarrow WL \quad \uparrow \text{reentry risk}$

How to  $\uparrow WL$ :  $\uparrow ERP/CV$  or both

$$\text{Wavelength}(WL) = CV \times ERP$$

(Distance) (speed) (time)

(tissue in refractory)

# An analogy

Length of the train (WL) vs Length of track  
(reentry circuit)

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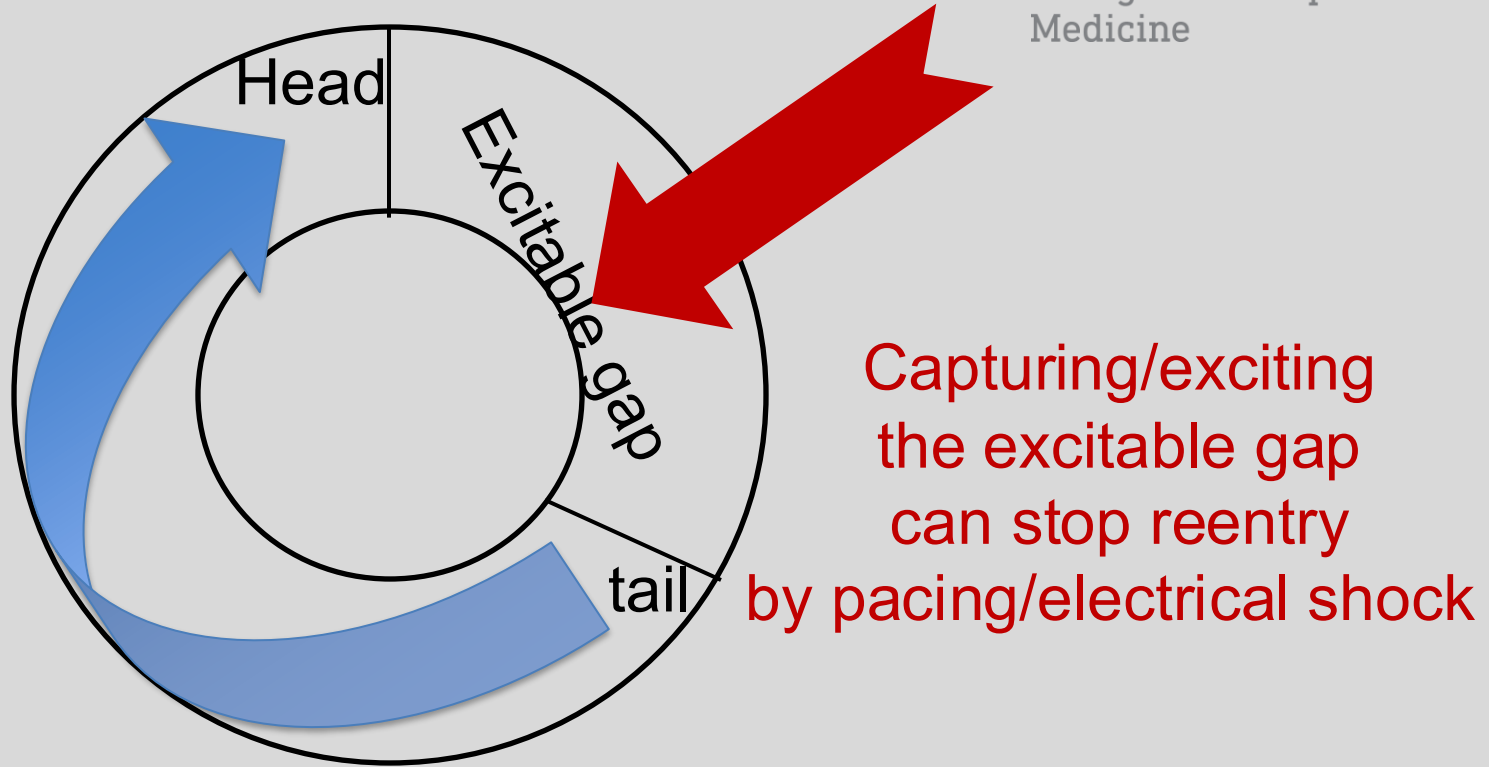


To play, the length of the train  $<$  length of the track  
On the other hand, if the train is very short, more trains could be on the same track

# Anti-tachycardia pacing (ATP)

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# Some examples of reentry circuit

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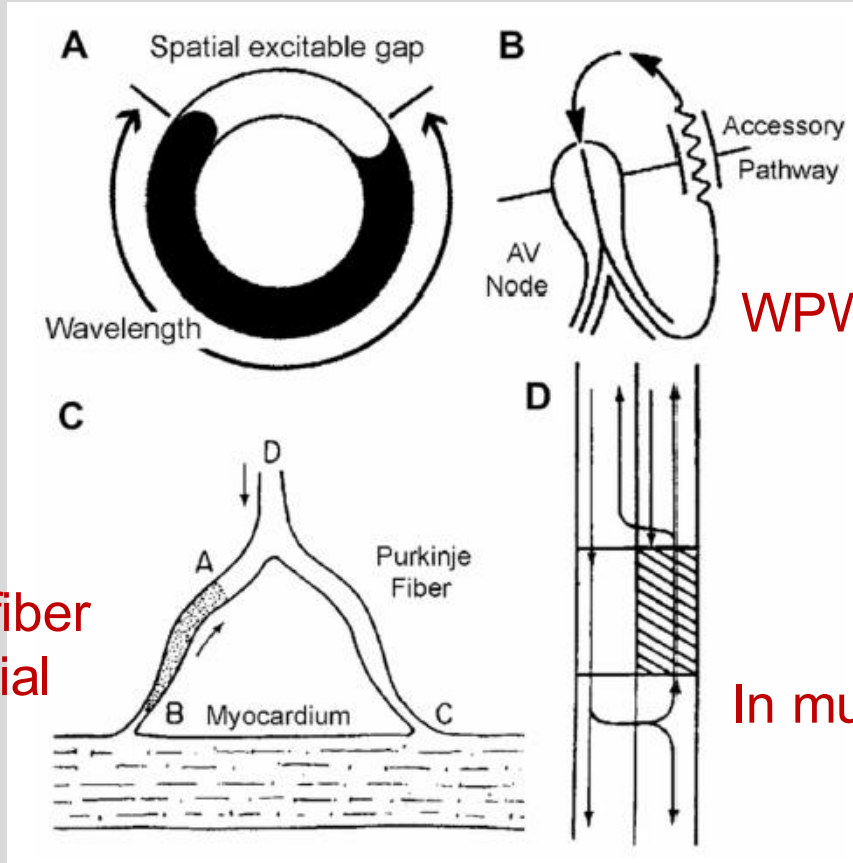
Macro-  
reentry

WPW syndrome

Micro-  
reentry

In muscle bundle

Purkinje fiber  
/myocardial



# Common clinical arrhythmias & mechanisms

## Bradyarrhythmias

1. Failure of impulse generation  
(decreased automaticity)
  - sinus bradycardia or arrest  
(Sick Sinus Syndrome, SSS)
2. Failure of impulse propagation  
(conduction block)
  - Sino-atrial block
  - AV block

# Tachy-arrhythmias

(1 or more of these 3 basic mechanisms: abnormal automaticity, TA, or reentry)

## 1. Supraventricular tachyarrhythmias

originates at His bundle or above, normal/narrow-QRS, (with the exception of preexisting abnormal ventricular conduction)

## 2. Ventricular arrhythmias: (below His bundle, wide-QRS)

# Supraventricular Tachyarrhythmias

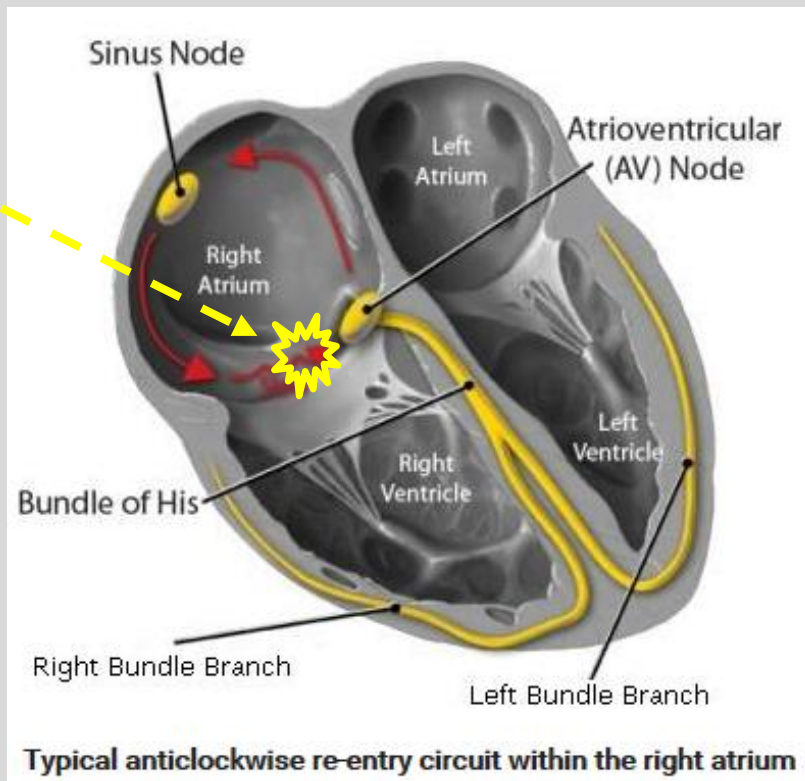
| <b>Clinical arrhythmia</b>                           | <b>Mechanisms</b>                                                |
|------------------------------------------------------|------------------------------------------------------------------|
| <b>Sinus tachycardia</b>                             | <b>Enhanced normal automaticity</b>                              |
| <b>Atrial premature contraction</b>                  | <b>Abnormal automaticity, TA, or reentry</b>                     |
| <b>Atrial tachycardia</b>                            | <b>Abnormal automaticity, TA, or reentry</b>                     |
| <b>Atrial flutter</b>                                | <b>Reentry</b>                                                   |
| <b>Atrial Fibrillation</b>                           | <b>Multiple reentries (ectopic foci + substrate)</b>             |
| <b>AVNRT</b> (AV nodal reentrant tachycardia)        | <b>Reentry</b>                                                   |
| <b>AVRT</b> (atrioventricular reentrant tachycardia) | <b>Reentry</b>                                                   |
| <b>Atrioventricular junctional tachycardia</b>       | <b>Enhanced normal automaticity, abnormal automaticity or TA</b> |

# Atrial flutter-Reentry in RA

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Ablation



IVC: inferior vena cava  
TV: tricuspid valve

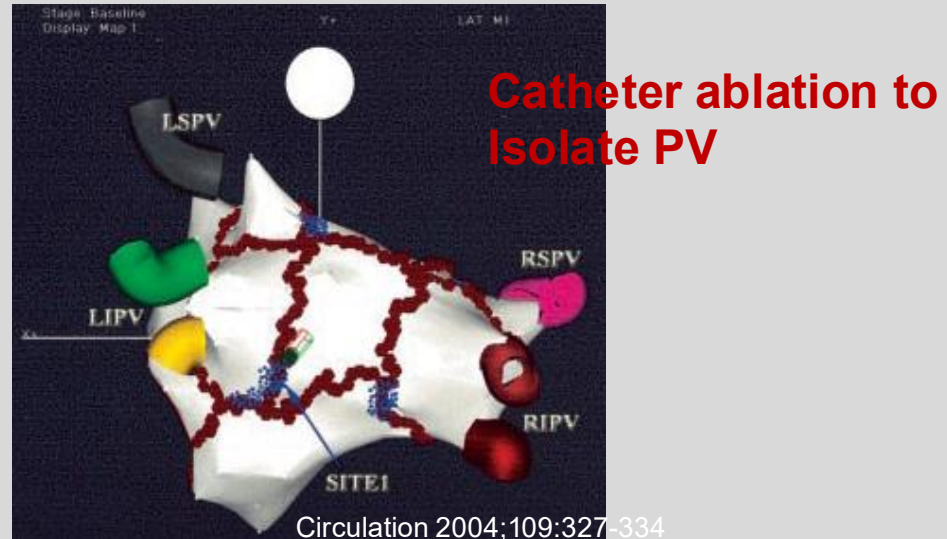
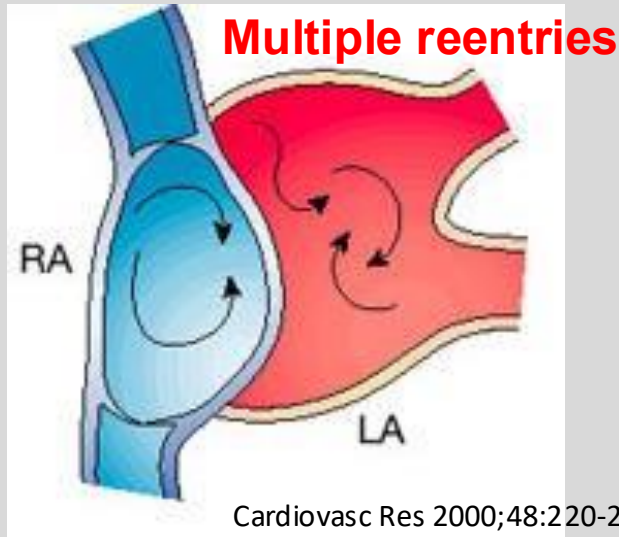
**Cavotricuspid isthmus:** between the IVC and the TV

# Atrial Fibrillation

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- The most common clinically significant arrhythmia
- **multiple reentries in both atria, requiring**
  - triggers or ectopic foci (abnormal automaticity, TA) and
  - substrate (remodeled atria, dilatation, fibrosis, to support multiple reentries).
- Most often triggers coming from the pulmonary veins (PV), that is why PV isolation is used to cure AF



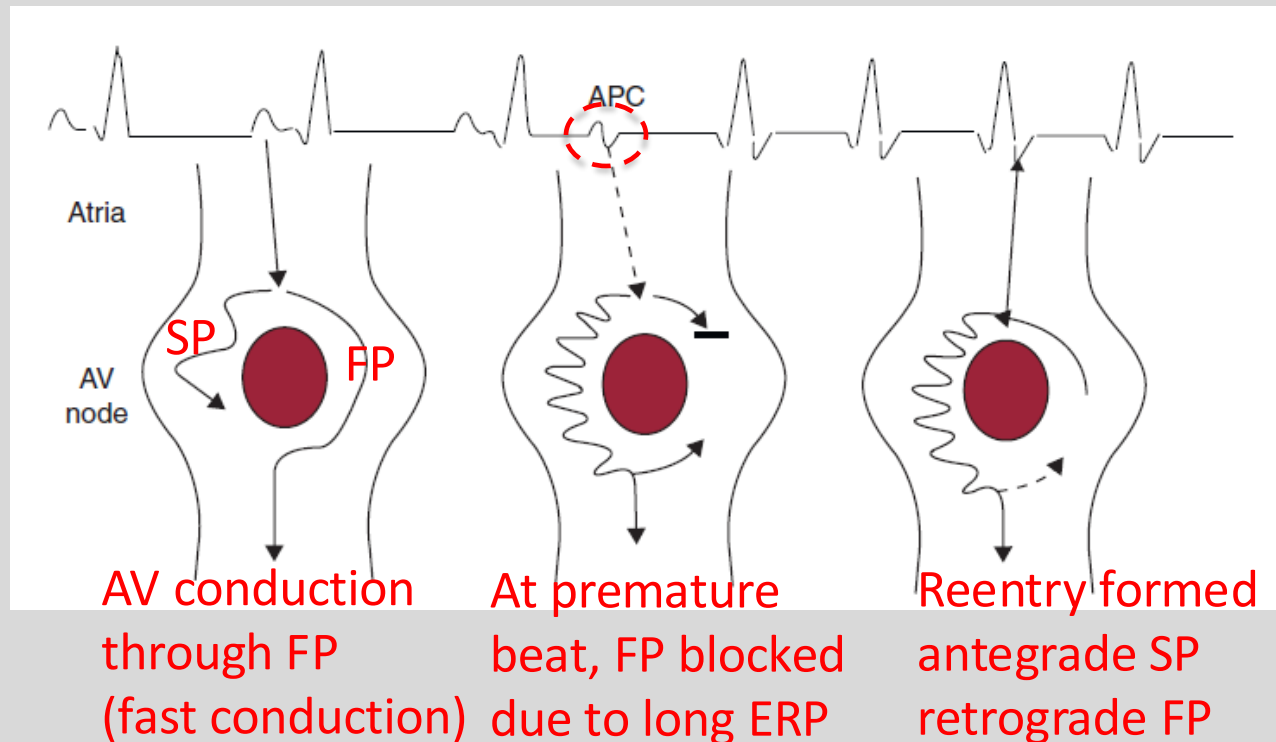
# AVNRT

## Atrioventricular Nodal Reentrant Tachycardia

reentry

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➤ AVN dual pathway electrophysiology

➤ SP ablation to cure AVNRT

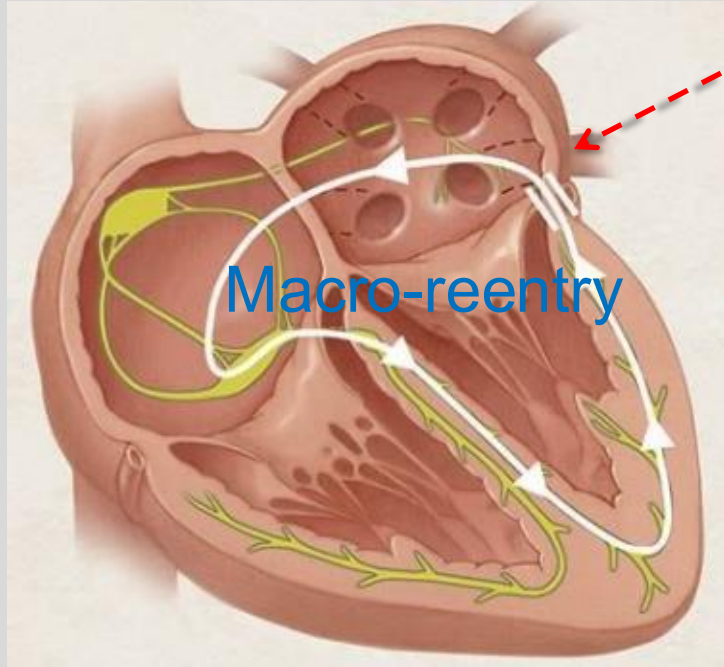
Micro-reentry

# AVRT

Atrioventricular Reentrant Tachycardia  
(Wolf-Parkinson-White (WPW) syndrome  
Or Pre-excitation Syndrome)

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Accessory pathway (**Bundle of Kent**)  
**Seen on ECG (WPW)**  
**concealed (no WPW on ECG)**  
**Some may have multiple pathways**  
**Cured by accessory pathway**  
**ablation**

If antegrade conduction through AVN-His-Purkinje system, then the tachycardia is with narrow-QRS.

If the reentry is in a reversed direction, then tachycardia is with widened-QRS.

# Minimum requirement

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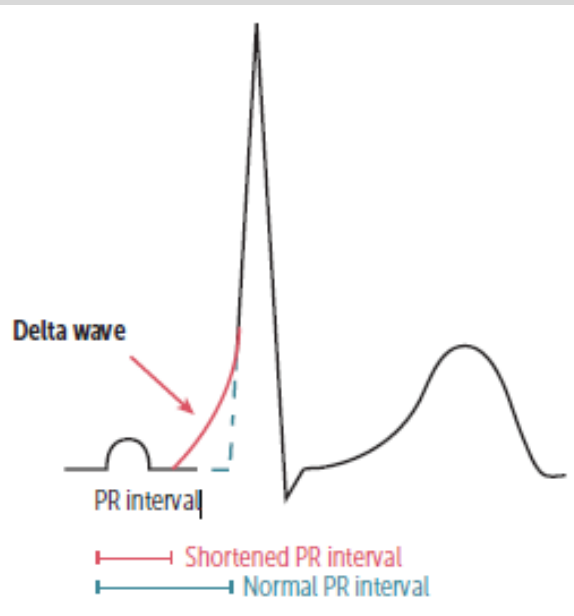
Wolff-Parkinson-White(WPW) syndrome:

Most common type of ventricular preexcitation syndrome.

Abnormal fast accessory conduction pathway from atria to ventricle (bundle of Kent) bypasses the rate-slowng AV node → ventricles begin to partially depolarize earlier → characteristic delta wave with widened QRS complex and shortened PR interval.

May result in reentry circuit

→ supraventricular tachycardia.



End of SVT  
Comes ...ventricular arrhythmias

# Ventricular tachyarrhythmias

(Abnormal, widened-QRS)

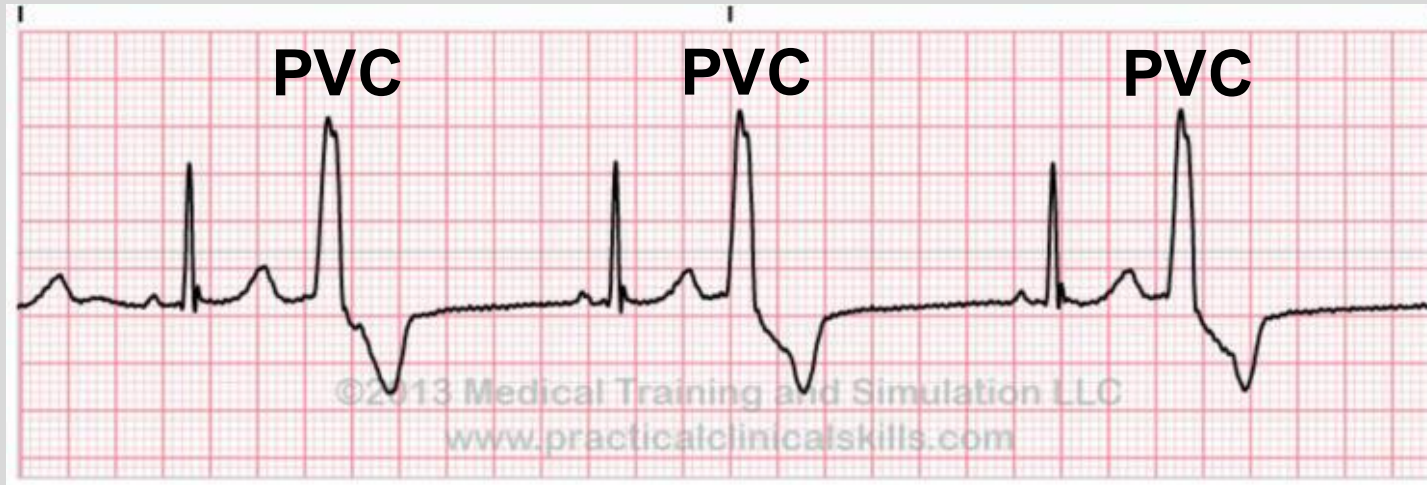
1. Premature ventricular contraction (PVC)
2. Ventricular tachycardia (VT)
3. Ventricular fibrillation (VF)

# 1. Premature ventricular contraction (PVC)

abnormal automaticity, TA or reentry

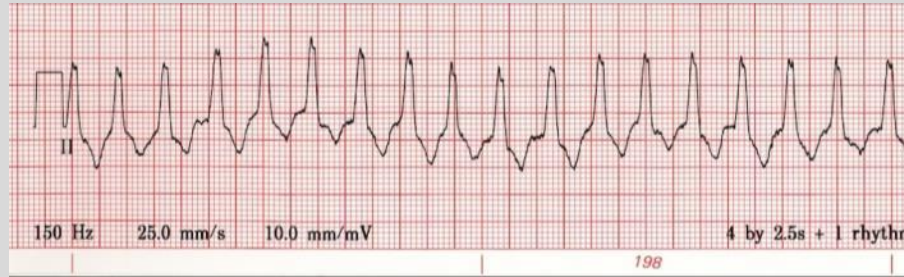
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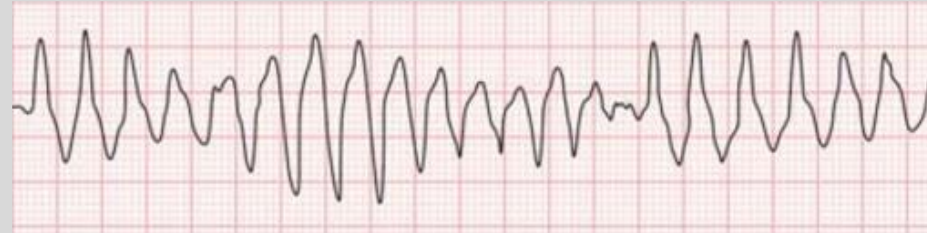
## 2. Ventricular tachycardia (VT)

### Monomorphic VT



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### Polymorphic VT

A specific type is termed Torsades de pointes, TDP

Typically seen in patients with  
long-QT syndrome  
or cardiac channelopathies

Brugada syndrome

Short QT syndrome

Catecholaminergic polymorphic VT  
(CPVT)

1. Structural heart disease  
ischemic, non-ischemic cardiomyopathy  
arrhythmogenic RV dysplasia  
**predominantly reentry**
2. Structurally normal hearts-idiopathic VT
  - a. Outflow tract tachycardia-**TA**
  - b. Fascicular ventricular tachycardia-**reentry**

# (1) Long QT syndrome

## a. Inherited (congenital) LQT syndrome

Reduced outward K<sup>+</sup> currents

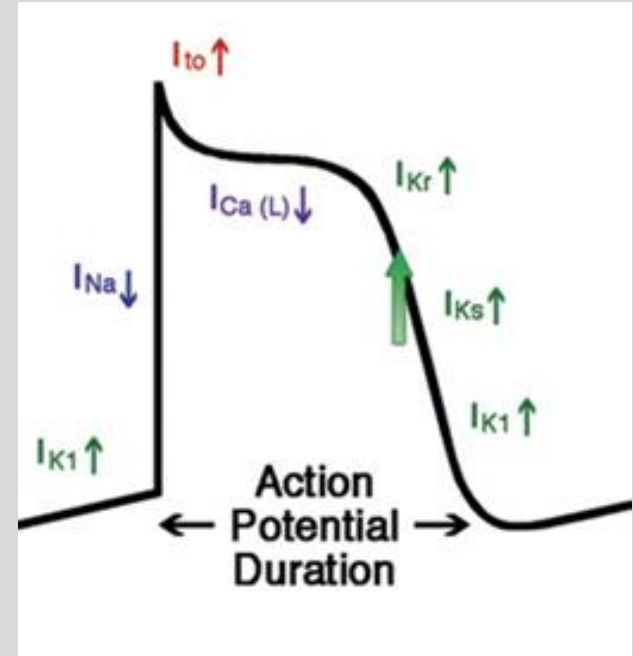
Increased inward Na<sup>+</sup> or Ca<sup>++</sup> currents

LQT1-IKs  
LQT2-Ikr } 3/4  
LQT3-INa  
LQT4,5....

TA (EAD) is the major mechanism involved in long-QT sympathetic excitation can increase inward currents and due to defects in outward channels, further prolonging the QT interval and increasing TDP. This is why  $\beta$ -blocker is typically used to prevent TDP in long-QT patients

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# Minimum requirement

## Congenital long QT syndrome

Inherited disorder of myocardial repolarization, typically due to ion channel defects; risk of sudden cardiac death (SCD) due to torsades de Pointes (**TDP-polymorphic VT**).

Includes:

**Romano-Ward syndrome**—autosomal dominant, pure cardiac phenotype (no deafness).

**Jervell and Lange-Nielsen syndrome**—autosomal recessive, sensorineural deafness.

## b. Acquired LQT syndrome

In some apparent healthy people  
when taking some medications or under  
certain conditions, they can develop  
Long-QT

These people may have some subtle  
genetic defects

# Common medications or conditions may lead to long QT

- Slow rate (bradycardia, complete heart block, etc.)
- Mechanical stretch
- Hypokalemia
- Hypoxia
- Acidosis
- Low extracellular  $K^+$  concentration
- Low extracellular  $Ca^{2+}$  concentration
- Low extracellular magnesium ( $Mg^{2+}$ ) concentration
- Class IA antiarrhythmic drugs (quinidine, disopyramide, procainamide)
- Class IB antiarrhythmic drugs (flecainide, encainide, indecainide)
- Class III antiarrhythmic drugs (amiodarone, sotalol, bretylium)
- Phenothiazines
- Tricyclic and tetracyclic antidepressants
- Erythromycin
- Antihistamines
- Cesium
- Amiloride
- Barium

# Minimum requirement

## Acquired LQT syndrome

Drug-induced long QT (**ABCDE**):

AntiArrhythmics (class IA, III)

AntiBiotics (e.g., macrolides)

Anti“C”ychotics (e.g., haloperidol)

AntiDepressants (e.g., TCAs)

AntiEmetics (e.g., ondansetron)

Polymorphic ventricular tachycardia (torsades de pointes, TDP), characterized by shifting sinusoidal waveforms on ECG; can progress to ventricular fibrillation.

Long QT interval predisposes to torsades de pointes.

Caused by **drugs**,  $\downarrow K^+$ ,  $\downarrow Mg^{2+}$ , other abnormalities.

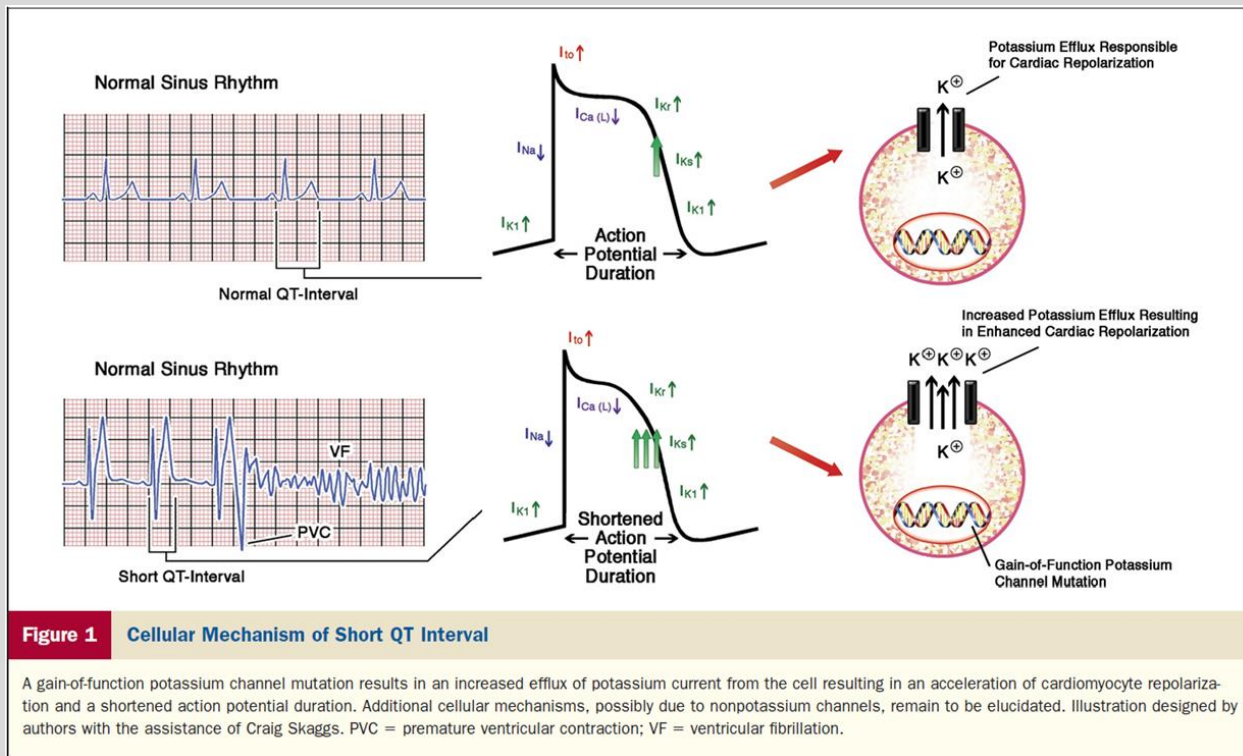
Treatment includes magnesium sulfate.

## (2) Short QT syndrome

Increased  $I_{K+}$ ,  $QT < 0.3s$

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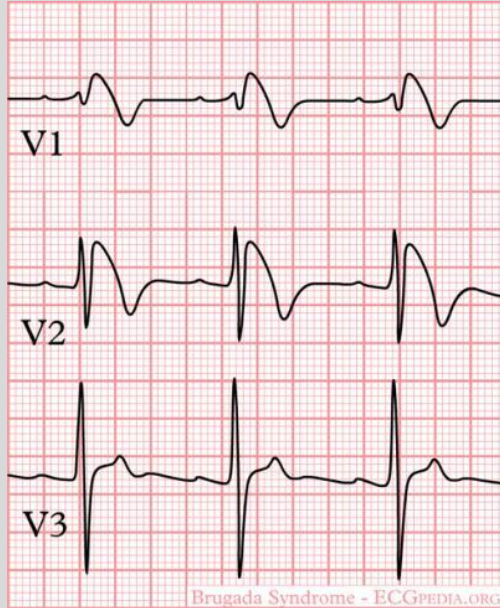
Short ERP  
Short WL  
Multiple reentries

# (3)Brugada syndrome

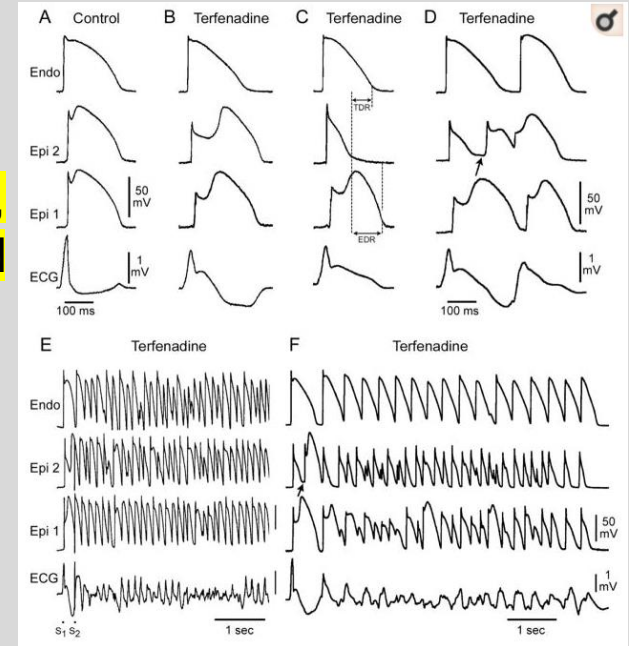
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diminished inward sodium current, particularly in RVOT(right  
ventricular outflow tract) epicardium-epicardial dispersion of  
repolarization



Some area  
repolarize earlier,  
can be re-excited  
by nearby  
myocytes



Typical ECG abnormalities in Brugada syndrome:  
a coved-type ST elevation in V1-V3

# Minimum requirement

## Brugada syndrome

Autosomal dominant disorder most common in Asian males.  
ECG pattern of pseudo-right bundle branch block and  
ST elevations in V1-V3.

Risk of ventricular tachyarrhythmias and SCD.

Prevent SCD with implantable cardioverter-defibrillator (ICD).

## (4) CPVT

Catecholaminergic polymorphic VT: due to leaky cardiac ryanodine receptor (RyR2)-**DAD**

genetic disorders of  $\text{Ca}^{++}$  release channels (RyR2) and their controlling proteins (e.g. **calquestrin**)

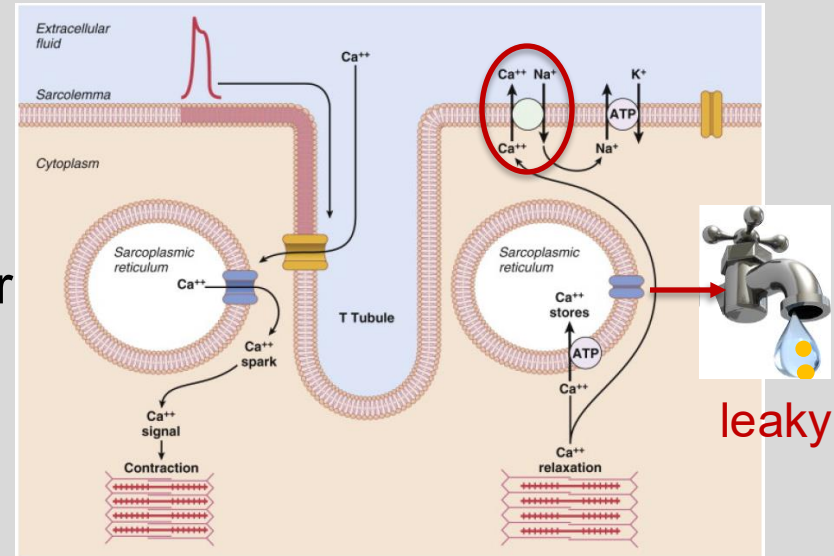
Typically seen in structurally normal heart

Sympathetic excitation like exercise or acute emotion could trigger CPVT

**$\beta$ -blocker therapy is used to prevent CPVT**

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## 4. Ventricular Fibrillation (VF)

initiated by any of the 3 mechanisms

-automaticity, TA and reentry, maintained by multiple wavelets, as in AF.

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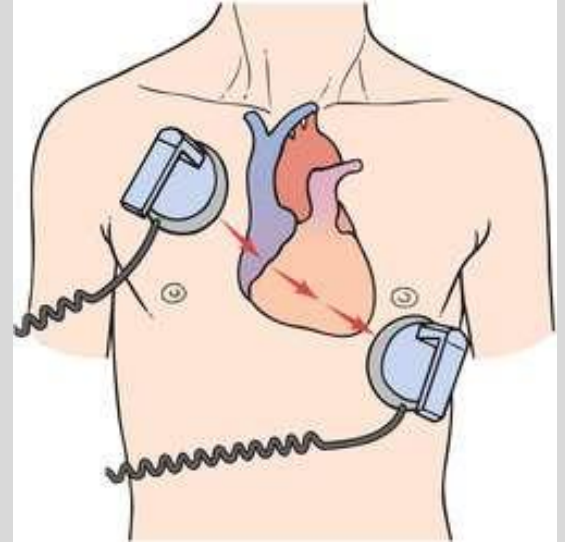
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VF=Loss of  
ventricular  
coordinated  
contraction



Patient is dying



Automated external defibrillator (AED) ---What's the mechanism?

Answer

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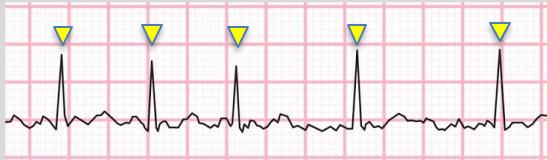
Electrical shock can excite  
all excitable gaps in multiple reentries

# Cardioversion vs Defibrillation

**Defibrillation:** VF, Synchronizer off, deliver shock immediately, require higher energy

**Cardioversion:** Other tachyarrhythmias, such as AF, PSVT or VT (if Sync is possible).

**Synchronizer on**  
(deliver on R wave,  
**NOT T wave**)

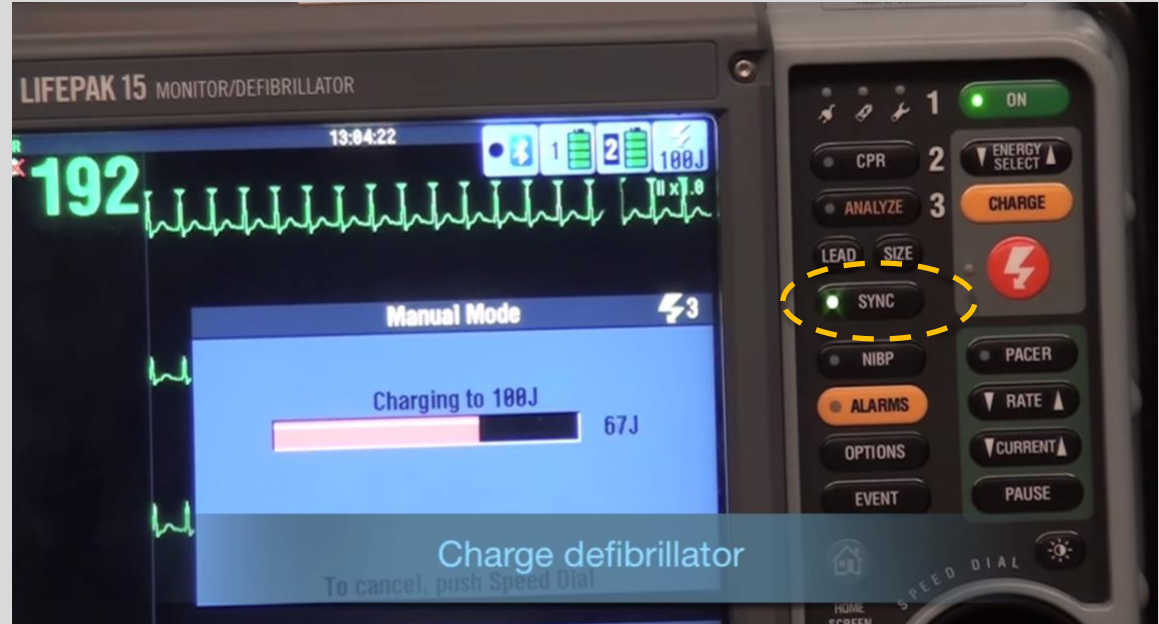


Use lower energy.

**Make sure not on T wave**  
**(why?)**

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Pictures from youtube.com

# Answer

Shock delivered on T wave  
can induce ventricular fibrillation  
(Electrophysiologically peak of T wave =  
vulnerable period of the ventricles)

Synchronize with peak of R wave, not T wave

# Summary Slide

- Common mechanisms of arrhythmias
- Formation of reentry, triggered activity, and abnormal automaticity
- Common factors affecting arrhythmia formation
- Common clinical arrhythmias and their underlying mechanisms
- Cardiac channelopathies and formation of arrhythmias
- Core References:
  - Guyton and Hall Textbook of Medical Physiology, 13e, Chapter 13
  - Journal article, Rev Esp Cardiol. 2012;65(2):174–185

# Lecture and Lab Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>